

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI, DUBAI CAMPUS
DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI
FIRST SEMESTER 2023 – 2024
DEPARTMENT OF COMPUTER SCIENCE

Questions: 6
Pages: 3

Question Paper

COURSE : Machine Learning (BITS F464)

COMPONENT : Midsem Exam CLOSED BOOK

WEIGHTAGE : 30% (30 Marks)

DATE : 3-November-2023

DURATION : 90 Minutes (8:30 am- 10:00 am)

Note:

1. Please write down any assumptions that you make.
2. Show the intermittant steps clearly for all answers.

Q. 1

The University of Nava Jeevan is working on classification model to classify the level of students into bright, average, and poor based on IQ test conducted for two subjects. The university received following confusion matrix after execution of prepared model.

Table 1: Confusion Matrix

Predicted values		Actual values		
		Bright	Average	Poor
	Bright	35	9	11
	Average	13	43	8
	Poor	9	10	24

Calculate the respective (individual) F1-score and accuracy for class bright, average and poor. Show all steps of calculation.

5 M

Q. 2

Consider a following dataset which shows willingness of certain customer to eat Veg or Nonveg Pizza in certain restaurant,

Table 2: Dataset of Willingness to eat

Restaurant	Type	Willingness to eat
Dominos	Veg	Yes
Dominos	Non-veg	Yes
Pizza hut	Veg	Yes
Pizza hut	Non-veg	No
Russo	Veg	No
Russo	Non-veg	No

- i. In above dataset, which feature (Restaurant or type) has the larger information gain with willingness to eat?
 - ii. Also draw the decision tree for predicting willingness to eat which maximizes the information gain?
- (Show all steps of calculation)

5 M

Q. 3

The road development authority of country wants to develop a classification to predict the accident based on weather condition, road condition, traffic condition, and engine problem. The data available with road development authority is as follows,

5 M

	<table><tr><th colspan="5">Table 3: Road development authority data</th></tr><tr><th>Weather Condition</th><th>Road Condition</th><th>Traffic Condition</th><th>Engine Problem</th><th>Accident</th></tr><tr><td>Rain</td><td>Bad</td><td>High</td><td>No</td><td>Yes</td></tr><tr><td>Snow</td><td>Average</td><td>Normal</td><td>Yes</td><td>Yes</td></tr><tr><td>Clear</td><td>Bad</td><td>Light</td><td>No</td><td>No</td></tr><tr><td>Clear</td><td>Good</td><td>Light</td><td>Yes</td><td>Yes</td></tr><tr><td>Snow</td><td>Good</td><td>Normal</td><td>No</td><td>No</td></tr><tr><td>Rain</td><td>Average</td><td>Light</td><td>No</td><td>No</td></tr><tr><td>Rain</td><td>Good</td><td>Normal</td><td>No</td><td>Yes</td></tr><tr><td>Snow</td><td>Bad</td><td>High</td><td>No</td><td>Yes</td></tr><tr><td>Clear</td><td>Good</td><td>High</td><td>Yes</td><td>No</td></tr><tr><td>Clear</td><td>Bad</td><td>High</td><td>Yes</td><td>Yes</td></tr></table> Identify the suitable classification technique for the available data to classify either accident will happen or not. Also identify the class for the test sample, X = (Weather Condition = Rain, Road Condition = Bad, Traffic Condition = Normal, Engine Problem = Yes).	Table 3: Road development authority data					Weather Condition	Road Condition	Traffic Condition	Engine Problem	Accident	Rain	Bad	High	No	Yes	Snow	Average	Normal	Yes	Yes	Clear	Bad	Light	No	No	Clear	Good	Light	Yes	Yes	Snow	Good	Normal	No	No	Rain	Average	Light	No	No	Rain	Good	Normal	No	Yes	Snow	Bad	High	No	Yes	Clear	Good	High	Yes	No	Clear	Bad	High	Yes	Yes	
Table 3: Road development authority data																																																														
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Q.4	Consider the following sample set shown in Table 4 consist of two independent features (IF) and one dependent feature (DF). Apply the Fisher linear discriminant analysis and plot the samples in reduced dimension. Show all steps of calculation. <table><tr><th colspan="3">Table 4: Sample Set</th></tr><tr><th>IF 1</th><th>IF 2</th><th>DF</th></tr><tr><td>40</td><td>36</td><td>Yes</td></tr><tr><td>11.1</td><td>37.2</td><td>Yes</td></tr><tr><td>30</td><td>36.5</td><td>Yes</td></tr><tr><td>21.4</td><td>39.4</td><td>Yes</td></tr><tr><td>10.7</td><td>39.6</td><td>Yes</td></tr><tr><td>3.4</td><td>40.7</td><td>Yes</td></tr><tr><td>42</td><td>37.6</td><td>No</td></tr><tr><td>31.1</td><td>42.2</td><td>No</td></tr><tr><td>50.0</td><td>38.5</td><td>No</td></tr><tr><td>60.4</td><td>39.4</td><td>No</td></tr><tr><td>45.7</td><td>38.6</td><td>No</td></tr><tr><td>17.3</td><td>42.7</td><td>No</td></tr></table>	Table 4: Sample Set			IF 1	IF 2	DF	40	36	Yes	11.1	37.2	Yes	30	36.5	Yes	21.4	39.4	Yes	10.7	39.6	Yes	3.4	40.7	Yes	42	37.6	No	31.1	42.2	No	50.0	38.5	No	60.4	39.4	No	45.7	38.6	No	17.3	42.7	No	8 M																		
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Q. 5	A biologist assumes that there is a linear relationship between the amount of fertilizer supplied to tomato plants and the subsequent yield of tomatoes obtained. Eight tomato plants of the same variety were selected at random and treated weekly with a solution in which x grams of fertilizer was dissolved in a fixed quantity of water. The yield y kilograms of tomatoes were recorded. <table><tr><th colspan="9">Table 5: Tomato plants data</th></tr><tr><th>Plant</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>G</th><th>H</th></tr><tr><td>x</td><td>1</td><td>1.5</td><td>2.0</td><td>2.5</td><td>3.0</td><td>3.5</td><td>4.0</td><td>4.5</td></tr><tr><td>y</td><td>3.9</td><td>4.4</td><td>5.8</td><td>6.6</td><td>7.0</td><td>7.1</td><td>7.3</td><td>7.7</td></tr></table> i. Calculate the linear regression equation and draw the best fit line. ii. Also calculate mean absolute error of training samples.	Table 5: Tomato plants data									Plant	A	B	C	D	E	F	G	H	x	1	1.5	2.0	2.5	3.0	3.5	4.0	4.5	y	3.9	4.4	5.8	6.6	7.0	7.1	7.3	7.7	4 M																								
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Q.6**3 M**

- i. For Ridge Regression, if the regularization parameter = 0, what does it mean?
(Select two correct options)
- (A) Large coefficients are not penalized
(B) Overfitting problems are not accounted for
(C) The loss function is as same as the ordinary least square loss function
(D) All of the above

- ii. In terms of the bias-variance trade-off, which of the following is/are substantially more harmful to the test error than the training error?
- (A) Bias
(B) Variance
(C) Loss
(D) Risk

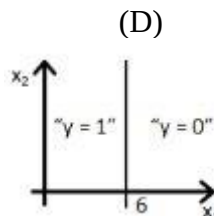
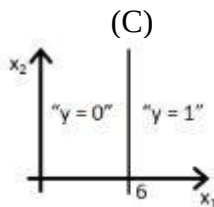
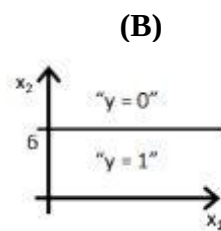
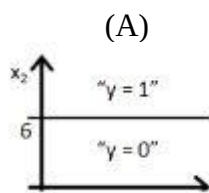
- iii. Suppose you train a logistic regression classifier and your hypothesis function H is

$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

Where

$$\theta_0 = 6, \theta_1 = 0, \theta_2 = -1.$$

Which of the following figure will represent the decision boundary as given by above classifier?



***** ALL THE BEST *****

Question Paper

COURSE : Machine Learning (BITS F464)

COMPONENT : Endsem Exam CLOSED BOOK

WEIGHTAGE : 40% (40 Marks)

DATE : 09-January-2024

DURATION : 3 hrs (8:30 am- 11:30 am)

Note:

1. Please write down any assumptions that you make.
2. Show the intermittant steps clearly for all answers.

Q1

Consider the following plotted sample data points, as shown in Figure 1.

8 M

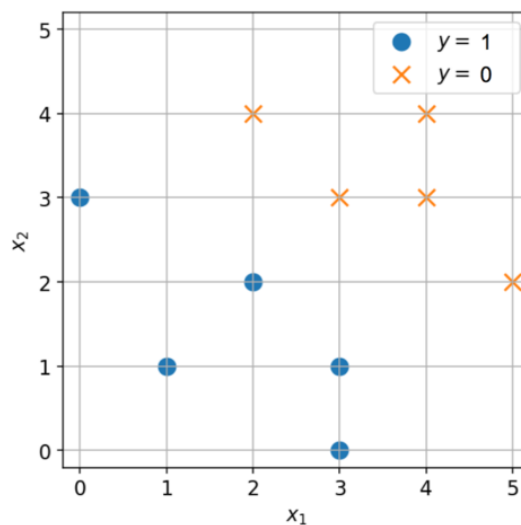


Figure 1: Sample Data Points

- a) Find the support vectors and draw the decision boundary for linear SVM. (Show all steps of calculating discriminating hyperplane) (6 M)
- b) Also calculate the size of margin. (2 M)

Q2

6 M

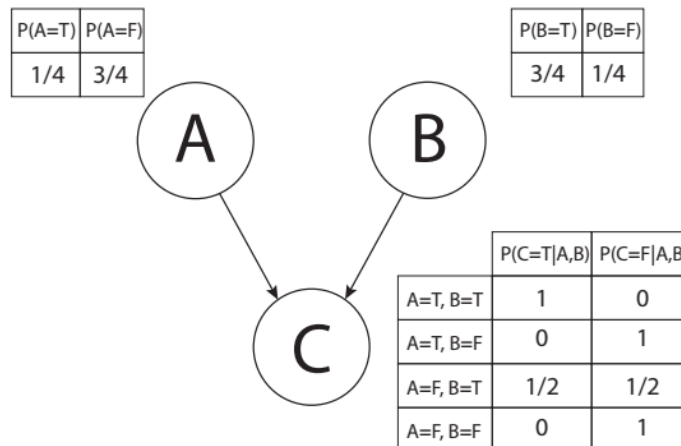
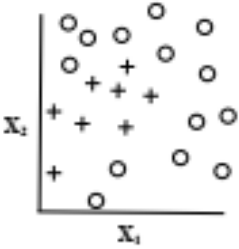
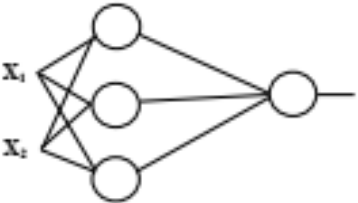
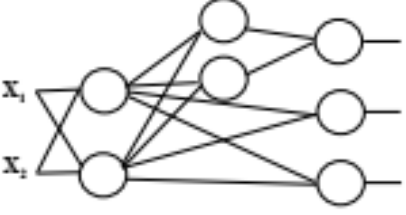
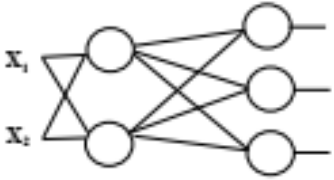
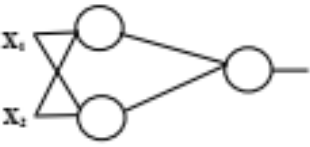


Figure 2: Bayesian net with probabilities

Compute the following probabilities by considering the above Bayesian net with probabilities shown in Figure 2,

	a) $P(C = T)$ b) $P(A = T, B = T)$ c) $P(A = T, B = T C = T)$	
Q3	You are given a design matrix $X = \begin{bmatrix} 2 & 1 \\ 4 & 5 \\ 4 & 3 \\ 6 & 6 \\ 6 & 8 \\ 7 & 8 \end{bmatrix}$. Use Principal Component Analysis (PCA) to reduce the dimension of matrices from 2 to 1. Show all steps of calculation of covariance matrix, eigen values, normalized eigen vectors, and sample points in one dimension with respect to Principal Component 1.	8 M
Q4	Define Markov Decision Process and consider a following Markov Decision Process (MDP) with rewards (R) for each state (as shown in Figure 3) and discount factor $\gamma = 0.8$, Figure 3: Markov Decision Process a) Calculate the accumulated rewards for path <i>Start</i> -> 2 -> 3 -> 4 -> <i>Start</i> (Show all steps of calculations) b) Find the optimum policy (policy with maximum accumulated reward) for moving from Start state to Exit state. (Show all steps of calculations)	6 M
Q5	Answer the following questions, a) Write any four correct differences between random forest and AdaBoost. b) In n-fold cross-validation each data point belongs to exactly one test fold, so the test folds are independent. Are the error estimates of the separate folds also independent? So, given that the data in test folds i and j are independent, are e_i and e_j, the error estimates on test folds i and j, also independent? c) How we calculate assignment score for each sample point in case of EM for Gaussian Mixture? And what indication assignment score of each sample points gives with respect to each Gaussian?	2 + 2 + 2 M

Q6	Recall that the output of a perceptron is 0 or 1. For each of the three following data sets, select the perceptron network with the fewest nodes that will separate the classes, and write the corresponding letter in the box. You can use the same network more than once.    A  B  C  D  E 	3 M
Q7	Answer the following multiple-choice question. Each question may have more than one correct option. 1) Large values of the log-likelihood statistic indicate: a. That there are a greater number of explained vs. unexplained observations. b. That the statistical model fits the data well. c. That as the predictor variable increases, the likelihood of the outcome occurring decreases. d. That the statistical model is a poor fit of the data. 2) Which of the following statements is not true about the ID3 algorithm? a. It is used to generate a decision tree from a dataset b. It begins with the original set S as the root node c. On each iteration of the algorithm, it iterates through every unused attribute of the set S and calculates the entropy or the information gain of that attribute d. Finally, it selects the attribute which has the largest entropy value 3) What's the bias and variance trade-off for KNN? a. A small 'k' results in a low bias but high variance (the model is sensitive to noise).	3 M

	b. A large 'k' results in a low bias but high variance (the model is sensitive to noise). c. A large 'k' leads to high bias but low variance (smoothing over the data). d. A small 'k' leads to high bias but low variance (smoothing over the data).	
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******* ALL THE BEST *******